

# Hosting

| Layer          | Choice                          | Why                                                                 |
|----------------|---------------------------------|---------------------------------------------------------------------|
| Compute        | AWS Mumbai (ap-south-1)         | Same region as Kite servers — sub-50ms latency, SEBI data residency |
| Hot DB (ticks) | QuestDB                         | Fastest tick ingestion, ASOF JOIN, time-series native               |
| Analytics DB   | DuckDB + Parquet                | Zero infra cost, columnar, plenty fast for our scale                |
| Object store   | S3 (Mumbai region)              | Backups, model artifacts, historical Parquet                        |
| Secrets        | AWS Secrets Manager             | Kite tokens, API keys, never on disk in plaintext                   |
| Monitoring     | Grafana + CloudWatch + Telegram | 3 layers: deep dive, native, instant alert                          |
| CI/CD          | GitHub Actions → AWS            | Auto-deploy from <code>main</code> , manual gate for production     |